Chapter
10

Medical Virology

Learning Objectives

After studying this chapter the students will be able to,

- *Study the importance of viruses causing disease in humans and from Animals. Viral infection are very common so its our main concern to protect us from the viral infection.*
- *Study the classification, structure and cultivation of Viruses.*
- *Study about the virus its pathogenesis, clinical feature and its treatment and prophylaxis.*
- *Study about the collection, processing of the sample and its molecular diagnosis.*

Chapter Outline

10.1 Evolutionary Origin of viruses

10.2 Cultivation of Viruses

10.3 Herpes Viruses

10.4 Hepatitis Viruses

10.5 Rabies Virus

10.6 Human Immuno Deficiency Virus

10.7 Arbo Virus

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Viruses are **infectious agents** with size ranging from about 20 nm to about 300 nm is diameter and contain only **one type of nucleic acid** (RNA or DNA) as their genome. The nucleic acid is encased in a protein shell, which may be surrounded by a lipid containing membrane. The entire infection unit is termed as **virion**. Viruses are obligate intracellular parasite, they replicate only is living cells. The study of virus is called as **virology**. **Martinus Beijerinck** is known as the Father of Virology.

10.1 Evolutionary Origin of Viruses

The origin of viruses is not known, but two theories of vital origin can be summarized as follows;

i. Viruses may be **derived from (DNA or RNA)** nucleic acid components of host cells to replicate and evolve independently.

ii. Viruses may be **degenerate forms** of intracellular parasites.

Morphology

Size

Viruses are smaller than bacteria, known as filterable viruses vary widely in size.

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